

Literature-Already-Answered Status Note

Enriched Kannan mappings and completeness

Status: literature already answered; not a new run result

Source question

Berinde and Păcurar introduce a self-map T of a normed space X as *enriched Kannan* when, for some $a \in [0, 1/2)$ and $k \geq 0$,

$$\|k(x - y) + Tx - Ty\| \leq a(\|x - Tx\| + \|y - Ty\|) \quad (x, y \in X).$$

On PDF page 5, item 3 of the remark following Theorem 2, they ask:

Does the enriched Kannan mapping principle still characterize the metric completeness of the ambient space?

This is the question in arXiv:1909.02379 [1]; the published version is identified as Remark 2.3 by the later paper. Since the displayed definition uses vector addition and scalar multiplication, the precise ambient category here is normed spaces.

Explicit later answer

Yu, Li, and Ji reproduce the same question as Question 1 on printed page 21582 (PDF page 3) and state on printed page 21583 that they give an affirmative answer [2]. Their Theorem 7 begins on that page, and its proof runs through printed page 21585 (PDF page 6). The proof starts from a nonconvergent Cauchy sequence in an incomplete normed space and constructs a fixed-point-free enriched Kannan self-map. Together with the complete-space fixed-point theorem of Berinde–Păcurar, this establishes the intended equivalence:

$$X \text{ is Banach} \iff \text{every enriched Kannan self-map } T : X \rightarrow X \text{ has a fixed point.}$$

The supporting authors explicitly know that they are answering the source question: their abstract names the Berinde–Păcurar question, and the main text labels it Question 1 immediately before presenting Theorem 7.

Statement qualification

The printed statement of Theorem 7 omits the universal fixed-point-property quantifier and literally says that if one map S has a fixed point, then the normed space is complete. That literal statement is false: a constant self-map is a Kannan, hence enriched Kannan, map with a fixed point on any nonempty incomplete normed space. The surrounding affirmative-answer sentence and the contradiction proof instead establish the universal characterization displayed above.

The intended reverse implication also follows immediately from the source definition: ordinary Kannan maps are precisely the $k = 0$ subclass of the enriched class. Hence the fixed-point property for all enriched Kannan maps implies Subrahmanyam’s fixed-point property for all ordinary Kannan maps, which characterizes metric completeness [3].

Classification and scope

The match is explicit and exact, so this packet belongs in `solutions/literature_already_answered/`. No part of the original normed-space completeness question remains open. This note does not claim the same formulation for arbitrary metric spaces without an added convex or linear structure.

The status search used the exact phrases “enriched Kannan mapping principle” and “characterizes completeness,” plus the source arXiv id. It located the 2024 AIMS article, whose abstract and Question 1 explicitly cite the original problem and whose Theorem 7 proof supplies the answer.

References

- [1] V. Berinde and M. Păcurar, *Fixed point theorems for Kannan type mappings with applications to split feasibility and variational inequality problems*, arXiv:1909.02379, 2019; published as *Kannan’s fixed point approximation for solving split feasibility and variational inequality problems*, J. Comput. Appl. Math. 386 (2021), 113217, doi:10.1016/j.cam.2020.113217.
- [2] Y. Yu, C. Li, and D. Ji, *Fixed point theorems for enriched Kannan-type mappings and application*, AIMS Mathematics 9(8) (2024), 21580–21595, doi:10.3934/math.20241048.
- [3] P. V. Subrahmanyam, *Completeness and fixed-points*, Monatsh. Math. 80 (1975), 325–330, doi:10.1007/BF01472580.